

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2021 INVITATIONAL B**

Biology

B01. A
B02. D
B03. A
B04. C
B05. B
B06. E
B07. B
B08. A
B09. A
B10. C
B11. E
B12. D
B13. B
B14. B
B15. E
B16. C
B17. C
B18. D
B19. D
B20. E

Chemistry

C01. A
C02. B
C03. D
C04. A
C05. C
C06. D
C07. D
C08. B
C09. E
C10. B
C11. E
C12. C
C13. A
C14. D
C15. A
C16. E
C17. B
C18. C
C19. B
C20. D

Physics

P01. C
P02. B
P03. E
P04. A
P05. A
P06. C
P07. E
P08. B
P09. B
P10. C
P11. E
P12. C
P13. A
P14. A
P15. E
P16. D
P17. B
P18. A
P19. C
P20. A

CHEMISTRY SOLUTIONS – UIL INVITATIONAL B 2021

- C01. (A) Mass of copper in a penny = $2.50 \text{ g} \times (2.5/100) = 0.0625 \text{ grams}$.
 $0.0625 \text{ g} / 63.55 \text{ g/mol} = 9.835 \times 10^{-4} \text{ moles Cu}$.
 $9.835 \times 10^{-4} \text{ moles} \times 6.022 \times 10^{23} \text{ atoms/mol} = 5.9 \times 10^{20} \text{ Cu atoms}$
- C02. (B) The fourth energy level is made up of *s*, *p*, *d*, and *f* subshells, and can therefore hold a maximum of $2 + 6 + 10 + 14 = 32$ electrons.
- C03. (D)
- C04. (A) The balanced equation for the reaction is $\text{CS}_2 + 3 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{SO}_2$
If 2.0 moles of CS_2 are used in the reaction, 6.0 moles of O_2 will be needed. Since the mass of the products is equal to the mass of the reactants, we can just calculate the total mass of the reactants and that will equal the total mass of the products.
 $(2.0 \text{ mol} \times 76.15 \text{ g/mol}) + (6 \text{ mol} \times 32.00 \text{ g/mol}) = 344.30 \text{ g}$.
- C05. (C) The reaction is endothermic, which means heat can be considered a reactant. Adding more heat will therefore shift the equilibrium to the right. Shaking the container will have no effect on the equilibrium. Reducing the volume will increase the pressure, and Le Chatelier's principle says the equilibrium will shift to alleviate the increased pressure. There are fewer moles of gas among the products than there are among the reactants, so the equilibrium will shift to the right to reduce the number of moles in the gas phase and therefore reduce the pressure.
- C06. (D) $1.645 \text{ g NaOH} / 40.00 \text{ g/mol} = 0.04113 \text{ mol NaOH}$. Need 0.04113 mol H^+ to neutralize this.
 $0.04113 \text{ mol H}^+ \times 1 \text{ mol H}_2\text{SO}_4 / 2 \text{ mol H}^+ = 0.02056 \text{ mol H}_2\text{SO}_4 \times (1 \text{ L} / 0.325 \text{ mol}) = 0.06327 \text{ L} = 63.3 \text{ mL}$
- C07. (D) The correct structure should have $5+6+6+6+1 = 24$ valence electrons, with octets on all four atoms. A violates the octet rule on N, B has too many electrons and violates the octet rule on N and one of the O atoms, C has too many electrons, D is correct, and E violates the octet rule on N.
- C08. (B) An oxidation-reduction reaction is one in which the oxidation numbers on two elements in the reaction change. Precipitation reactions and acid-base reactions do not involve changes in the oxidation states of the atoms – they just involve moving an ion from one species to another – but all other reaction types do.
- C09. (E) The only monatomic gas at room temperature with an atomic mass close to 39.96 is argon, which has 18 protons. Since the atom has a mass of almost exactly 40, it must have a mass number of 40. It is therefore an argon isotope with 22 neutrons (the most common and most stable isotope of argon).
- C10. (B) Larger atoms with more electrons are more easily polarizable, resulting in more dispersion forces between atoms and therefore a higher melting point and a higher boiling point. All of the other answer choices are false statements.

- C11. (E) Initial volume = $nRT/P = (2.5)(0.08206)(85+273)/1 = 73.44 \text{ L}$
 Final volume = $nRT/P = (2.5)(0.08206)(-40+273)/0.60 = 79.67 \text{ L}$
 $\Delta V = V_{\text{final}} - V_{\text{initial}} = 79.67 - 73.44 = 6.2 \text{ L}$
- C12. (C) $\Delta G = \Delta H - T\Delta S$, and a process is spontaneous when ΔG is negative. If the process is endothermic with an increase in entropy, ΔH and ΔS are both positive, so for ΔG to be negative, $T\Delta S$ must be larger than ΔH . The process is therefore energetically more favorable at higher temperatures. Melting ice and boiling water are both examples of this kind of physical transformation.
- C13. (A) For a compound with 3:2 ion ratio, $K_{\text{sp}} = 108x^5$ where x is the molar solubility of the compound. $2.2 \times 10^{-33} = 108x^5$, so $x = 1.153 \times 10^{-7} \text{ M}$. Since one mole of $\text{Cd}_3(\text{AsO}_4)_2$ produces 3 moles of Cd^{2+} ions, $[\text{Cd}^{2+}] = 3 \times 1.153 \times 10^{-7} \text{ M} = 3.5 \times 10^{-7} \text{ M}$.
- C14. (D) A catalyst is consumed early in a reaction mechanism and is produced later in the mechanism, and does not appear in the overall equation for the chemical reaction.
- C15. (A) work = $-P\Delta V = -(1)(1.7 - 3.0) = 1.3 \text{ L}\cdot\text{atm} \times (101.325 \text{ J})/(1 \text{ L}\cdot\text{atm}) = 132 \text{ J}$
- C16. (E) $10.0 \text{ g Na}/22.99 \text{ g/mol} = 0.43497 \text{ moles of Na}$, enough to make 0.43497 moles of NaCl
 $PV = nRT$, $n = PV/RT$, $n = (5.0 \text{ L Cl}_2)(1 \text{ atm})/(0.08206)(273) = 0.22319 \text{ moles of Cl}_2$, enough to make 0.44638 moles of NaCl.
 Na is the limiting reactant. $0.43497 \text{ moles NaCl} \times 58.44 \text{ g/mol} = 25.4 \text{ grams of NaCl}$.
- C17. (B) $7.6 \text{ g AgCl} / 143.32 \text{ g/mol} = 0.05303 \text{ moles AgCl}$, so there must have been 0.05303 moles of Ag^+ in the solution. $0.05303 \text{ mol} / 0.1000 \text{ L} = 0.53 \text{ M}$.
- C18. (C) Based on its position in the periodic table, copper should have an electron configuration ending in $4s^2 3d^9$, but bumping one $4s$ electron into the $3d$ subshell fills the $3d$ subshell and results in a more stable electron configuration of $4s^1 3d^{10}$.
- C19. (B) HClO has the largest K_{a} , therefore since the concentrations are all equal it will have the lowest pH. Assuming $[\text{H}^+] \ll C_{\text{ACID}}$,
- _____
- The assumption checks out, $5.9 \times 10^{-5} \ll 0.10$
 $\text{pH} = -\log[\text{H}^+] = -\log(5.9 \times 10^{-5}) = 4.23$.
 (Hypopotamous acid and HIP_2O are not real. I made those up.)
- C20. (D) $128.56 \text{ g} / 142.05 \text{ g/mol} = 0.90503 \text{ moles Na}_2\text{SO}_4$. $0.90503 \text{ moles} / 0.500 \text{ mol/L} = 1.81 \text{ L}$.

PHYSICS SOLUTIONS – UIL INVITATIONAL B 2021

- P01. (C) page 9: “In short, the theory describes a colourful and amazing world where universes explode, space collapses into bottomless holes, time sags and slows near a planet, and the unbounded extensions of interstellar space ripple and sway like the surface of the sea...”
- P02. (B) page 12: “It was to be Einstein once again who, five years later, came to understand that the ‘packets of energy’ were real. Einstein showed that light is made of packets: particles of light. Today we call these ‘photons’.”
- P03. (E) page 23: “Soon someone (perhaps Parmenides, perhaps Pythagoras) realized that the sphere is the most reasonable shape for this flying Earth for which all directions are equal – and Aristotle devised convincing scientific arguments to confirm the spherical nature of both the Earth and of the heavens around it...”
- P04. (A) H-II regions consist of about 90% hydrogen. This hydrogen is heated by nearby hot stars which excites the hydrogen atoms into various energy states. The excited atoms then decay and emit photons of specific wavelengths. These wavelengths are grouped together in named series – Lyman, Balmer, Paschen, Brackett, Pfund... The Balmer series of wavelengths are the only ones in the visible portion of the spectrum, and the strongest Balmer line has a wavelength of 656nm – which is red. Because this visible spectral line is so strong, H-II regions generally exhibit a red color.
- P05. (A) First, we must consider the subtraction in the numerator. These both are multiplied by the same power of ten, so we do not need to make any adjustments for that. Also, the power of ten is irrelevant for determining the significant figures.
When adding or subtracting, the correct number of significant figures is based on the value with the least decimal places. The first value, 1.98, has two decimal places. The second value, 2.002, has three decimal places. Therefore, the result of the subtraction would have significant figures to only two decimal places. The result is: $[1.98] - [2.002] = -[0.02]2$. Stopping at two decimal places means that this result has only a single significant figure. The complete result, including the power of ten, is $-[0.02]2 \times 10^3 = -[2]2$.
Now we consider the division. When dividing or multiplying, the value with the least number of significant figures determines the number of significant figures of the result. For our case:
 $\frac{-[2]2}{[19.3]} = -1.139896373$. The numerator has one significant figure, and the denominator has three.
Thus, the answer will have only one significant figure: $-[1].139896373 \approx -1$.
- P06. (C) We know the initial velocity, the acceleration (g), and the displacement; and we are looking for the final velocity. The best kinematic equation to use is $v_f^2 = v_i^2 + 2a\Delta x$. Taking downward to be negative, and putting in the known values, we get: $v_f^2 = (-5.25)^2 + 2(-9.8)(-2.29)$. This gives: $v_f^2 = 72.45 \rightarrow v_f = 8.51\text{m/s}$.
- P07. (E) In the free-body diagram, each block has two forces acting on it – the weight, directed downward, and the tension, directed upward. In addition, each block accelerates, with the heavier block accelerating downward, and the lighter block accelerating upward. Taking upward to be positive, and summing up the forces on each block, we get two equations. For the heavier block, which accelerates downward: $\sum F_1 = T - (4.2)(9.8) = (4.2)(-a) \rightarrow T - 41.16 = -4.2a$. For the lighter block, we get: $\sum F_2 = T - (2.8)(9.8) = (2.8)(a) \rightarrow T - 27.44 = 2.8a$.

Solving for tension in the second equation: $T = 2.8a + 27.44$, and substituting this into the first equation gives the acceleration: $(2.8a + 27.44) - 41.16 = -4.2a \rightarrow -13.72 = -7.0a$. So, the acceleration is $a = 1.96 \text{ m/s}^2$. Then, we can easily find the final velocity: $v_f = v_i + at = 0 + (1.96)(1.30) = 2.55\text{m/s}$.

- P08. (B) This can only be solved by considering conservation of momentum. Because we have no external forces, like friction, the total momentum before the collision will equal the total momentum after the collision. For these calculations, we will take “to the right” to be positive, and “to the left” to be negative. Before the collision, the total momentum is $p_i = m_r v_{ri} + m_g v_{gi}$. Initially, the green block isn’t moving, so $v_{gi} = 0$. Putting in what we know about the initial velocity and mass of the red block, we get: $p_i = (0.300\text{kg})(5.20\text{m/s}) = 1.56\text{ kgm/s}$.
- After the collision, both blocks are moving, so we have a final momentum of $p_f = m_r v_{rf} + m_g v_{gf}$. This equals the initial momentum, $m_r v_{rf} + m_g v_{gf} = 1.56\text{ kgm/s}$. Putting in the known values (and being careful with signs), we get: $(0.300\text{kg})(-0.850\text{m/s}) + (0.420\text{kg})v_{gf} = 1.56\text{ kgm/s}$. This leads to: $-0.255 + 0.420v_{gf} = 1.56 \rightarrow 0.420v_{gf} = 1.815 \rightarrow v_{gf} = 4.32\text{ m/s}$.
- P09. (B) We begin by considering the forces acting on the block, of which there are two: tension, T , directed upward, and weight, $m_b g$, directed downward. The block is also accelerating downward. Putting this into Newton’s acceleration law: $\sum F = T - m_b g = m_b (-a) = -m_b a$. Putting in the known values gives: $T - (12)(9.8) = -(12)a = T - 117.6 = -12a$.
- Now we consider the torque acting on the pulley and the rotational acceleration of the pulley. There is only one force that contributes to the rotational acceleration of the pulley: the tension, T . Mathematically, $\sum \tau = Tr = I\alpha$. Putting in the known values: $T(0.30) = (0.220)\alpha$.
- Since the rope unwinds without slipping, we know that the angular acceleration of the pulley is related to the linear acceleration of the mass: $a = r\alpha$, which gives: $a = (0.30)\alpha$, or $\alpha = 3.33a$.
- From these last two equations: $T = 0.733\alpha = 2.442a$. Plugging this into the first equation, we get: $T - 117.6 = 2.442a - 117.6 = -12a \rightarrow 14.442a = 117.6 \rightarrow a = 8.14\text{ m/s}^2$.
- P10. (C) The frequencies resonant in an open-closed pipe are given by the equation $f_n = \frac{nv}{4L}$ (odd n). At an air temperature of 20°C , the speed of sound is $v = 343\text{ m/s}$. And for the fundamental frequency, $n = 1$. Putting this all together gives: $280 = \frac{(1)(343)}{4L} \rightarrow L = \frac{343}{(4)(280)} = 0.31\text{m} = 31\text{cm}$.
- P11. (E) The equation describing linear thermal expansion is $\Delta L = \alpha L_0 \Delta T$. Between summer and winter, the temperature changes by $|\Delta T| = |45 - (-10)| = 55^\circ\text{C}$. Thus, the contraction of each bridge section would be $\Delta L = \alpha L_0 \Delta T = (1.0 \times 10^{-5})(12.5)(55) = 0.006875\text{m}$. Since each section contracts by this amount, this is the amount by which each gap will widen. The gaps start at 0.50cm , and widen by 0.6875cm , so the width of each gap in the winter is $0.50 + 0.6875 = 1.1875 \approx 1.2\text{cm}$.
- P12. (C) In order to find the current flowing in the one resistor, we must first find the total equivalent resistance of the circuit. We begin by combining the 1200Ω and 1000Ω in series: $R_A = 1200 + 1000 = 2200\Omega$. Now, this equivalent resistance is combined in parallel with the 1800Ω resistor: $\frac{1}{R_B} = \frac{1}{1800} + \frac{1}{2200} \rightarrow R_B = 990\Omega$. Finally, we combine this group resistance in series with the 510Ω resistor: $R_T = 510 + 990 = 1500\Omega$. This is the total equivalent resistance of the circuit.
- From here, we can determine the current flowing from the battery: $I_T = \frac{V_T}{R_T} = \frac{24.0\text{V}}{1500\Omega} = 0.016\text{A}$. Because they are in series, this same current flows through the 510Ω resistor and the 990Ω equivalent resistance ($I_B = I_T$). This allows us to determine the voltage across the 990Ω group: $V_B = I_B R_B = (0.016)(990) = 15.84\text{ V}$. Because they are in parallel, the voltage across the 1800Ω resistor and the voltage across the 2200Ω equivalent resistance are equal to the voltage across the group. This allows us to finally determine the current flowing through the 1800Ω resistor: $I_{1800} = \frac{V_{1800}}{R_{1800}} = \frac{15.84}{1800} = 0.0088\text{A} = 8.8\text{mA}$.

- P13. (A) Coulomb's Law gives the force between two charges, but we will have to work one at a time with the fixed charges. First, let's consider the force on the $+0.50\mu\text{C}$ charge at the origin due to the $1.20\mu\text{C}$ charge: $F_1 = \frac{kQ_1Q_0}{r_{10}^2} = \frac{(8.99 \times 10^9)(1.20 \times 10^{-6})(0.50 \times 10^{-6})}{(0.01)^2} = 53.94 \text{ N}$. Since these are both positive charges, the force will be repulsive. Thus, this force on the $+0.50\mu\text{C}$ charge is directed to the right. Now, we can consider the force on the $+0.50\mu\text{C}$ charge at the origin due to the $-1.55\mu\text{C}$ charge: $F_2 = \frac{kQ_2Q_0}{r_{20}^2} = \frac{(8.99 \times 10^9)|-1.55 \times 10^{-6}|(0.50 \times 10^{-6})}{(0.02)^2} = 17.42 \text{ N}$. Since these charges are opposite in sign, the force will be attractive. Therefore, this force on the $+0.50\mu\text{C}$ charge will also be directed to the right. The total force is the combination of these two. Since both forces are directed to the right, they add together to give the total: $F_T = F_1 + F_2 = 53.94 + 17.42 = 71.4 \text{ N}$.
- P14. (A) First, we need to find the force on the projectile. Since the current and the magnetic field are perpendicular, we can use the simple formula: $F = I\ell B = (60\text{A})(0.25\text{m})(0.85\text{T}) = 12.75 \text{ N}$. Now, we can use Newton's second law to calculate the acceleration: $a = \frac{F}{m} = \frac{12.75\text{N}}{0.150\text{kg}} = 85.0 \text{ m/s}^2$. Finally, we can use kinematics to determine the velocity after the projectile accelerates down the length of the rails: $v^2 = v_0^2 + 2a\Delta x = 0 + 2(85.0)(3.50) = 595 \rightarrow v = 24.4 \text{ m/s}$.
- P15. (E) Light reflected from a surface is maximally polarized when the light reflects at Brewster's angle. So, based on this, and on what is given in the problem, we know that Brewster's angle is $\theta_p = 61.0^\circ$. Brewster's angle is related to the index of refraction of the surface by the equation $\frac{n_{\text{surface}}}{n_{\text{medium}}} = \tan\theta_p$. The medium through which the light is travelling is air, so $n_{\text{medium}} = 1$. Thus, the index of refraction of the surface is: $n_{\text{surface}} = \tan(61.0^\circ) = 1.80$.
- P16. (D) Based on the size of the letters on the coin, we can determine the magnification provided by the lens: $M = \frac{h'}{h} = \frac{7.00\text{mm}}{3.00\text{mm}} = 2.33$. Now, we can relate this magnification to the image and object locations (and we know the object location, $p = 4.00\text{cm}$.) $M = 2.33 = -\frac{q}{p} = -\frac{q}{4.00\text{cm}}$. This allows us to determine the image location: $q = -2.33(4.00) = -9.33\text{cm}$. This is negative because the image is virtual. Finally, we can use the image and object locations to calculate the focal length of the lens: $\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{1}{4.00\text{cm}} + \frac{1}{-9.33\text{cm}} = 0.143 \rightarrow f = 7.00\text{cm}$.
- P17. (B) The energy/time version of Heisenberg's uncertainty principle is $\Delta E \Delta t \geq \frac{\hbar}{2}$. We can use the equality in this equation since we are given the minimum rest-mass energy uncertainty. Now, we must deal with units and constants:
First, $24.0\text{MeV} = (24.0 \times 10^6) * (1.602 \times 10^{-19}) = 3.84 \times 10^{-12} \text{ J}$.
Also, $\frac{\hbar}{2} = \frac{h}{4\pi} = \frac{6.626 \times 10^{-34}}{4\pi} = 5.27 \times 10^{-35} \text{ Js}$. Putting it all together, we get:
 $(3.84 \times 10^{-12})\Delta t = 5.27 \times 10^{-35}$, which gives $\Delta t = 1.37 \times 10^{-23} \approx 10^{-23} \text{ seconds}$. This is the approximate lifetime of the subatomic particle.
- P18. (A) The equation that describes radioactive decay is $N = N_0 e^{-\lambda t}$. Here λ is the decay constant and is related to the half-life by $\lambda = \frac{\ln(2)}{T_{1/2}} = \frac{0.693}{700 \text{ million years}} = 9.90 \times 10^{-10} \text{ year}^{-1}$. Putting this, and the other given values, into the decay equation gives:
 $N = (200.0\text{g})e^{-(9.90 \times 10^{-10})(280 \times 10^6)} = (200.0\text{g})e^{-0.277} = 152\text{g}$.

- P19. (C) The equation describing the period of a pendulum is $T = 2\pi\sqrt{\frac{\ell}{g}}$ where ℓ is the length of the pendulum and g is the acceleration due to gravity. Squaring both sides to get the period-squared gives: $T^2 = \frac{4\pi^2}{g}\ell$. We can see a plot of T^2 versus ℓ would be linear with a slope of $\frac{4\pi^2}{g}$. Turning our attention to the graph given, we can find the slope of the line. We need two points on the line, for which I choose (20.0cm, 1.00sec²) and (50.0cm, 2.45sec²). Then the slope of the line is: $slope = \frac{2.45-1.00}{50.0-20.0} = \frac{1.45}{30} = 0.0483$. Setting this equal to the slope formula from the equation gives us: $0.0483 = \frac{4\pi^2}{g} \rightarrow g = \frac{4\pi^2}{0.0483} = 816.8 \text{ cm/s}^2$. Converting to meters gives: $g = 8.168 \approx 8.2 \text{ m/s}^2$.
- P20. (A) The relationship between the electric field and the electric potential can be summed up as $|E| \propto \frac{\Delta V}{\Delta x}$. This basically means that the electric field is greatest at places where the electric potential changes rapidly over a short distance. On a diagram like the one given, the strongest electric fields are located where the equipotential surfaces are spaced close together. In comparing the different points, the equipotential surfaces are closest together at point A. Thus, the greatest electric field strength is found at point A.